Engineering-Prior-Guided NSGA-II for Elastic Capacity-Load Ratio Planning under High Distributed Photovoltaic Penetration

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***Abstract*—High-penetration distributed PV challenges fixed capacity-load ratio planning in county 110 kV grids by introducing reverse power and changing N-1 supply risk. This paper formulates an elastic per-station planning model that couples capacity-load ratio, storage power, reverse-power constraints, and supply risk. Engineering-Prior-Guided NSGA-II (EPG-NSGA-II) combines feasibility-aware initialization with repair projection to improve constrained search efficiency. In a synthetic county case with K=20, the selected elastic configuration reduces the annualized cost index by 13.1% relative to the uniform R=2.0 baseline under equal EENS. At K=40, EPG-NSGA-II reaches a comparable hypervolume with fewer evaluations than plain NSGA-II.**

***Index Terms*—Distributed photovoltaics, capacity-load ratio, NSGA-II, multi-objective optimization, storage planning.**

I. INTRODUCTION

High-penetration distributed PV is making county 110 kV grids increasingly bidirectional. Midday PV output may exceed local demand, causing reverse power and changing the N-1 supply margin [3]. A uniform county-wide capacity-load ratio may over-invest at moderate-PV stations while failing to reflect reverse-power pressure at PV-dense stations [1], [2]. Existing planning often relies on rigid ratio rules, while evolutionary optimization can search the cost-risk trade-off directly [1], [2], [5], [6]. Under strong engineering constraints, random initialization and variation can generate many infeasible individuals.

This paper proposes EPG-NSGA-II to address this feasibility bottleneck. It keeps nondominated sorting unchanged and adds two engineering-prior operators: warm-start sampling within engineering bounds and repair projection after variation. The contributions are threefold: an elastic capacity-load ratio model with reverse-power and N-1 constraints, an EPG-NSGA-II strategy with warm-start sampling and repair projection, and a synthetic county case with ablation analysis.

II. ENGINEERING-PRIOR-GUIDED NSGA-II

A. Planning Model

The model considers a set of 110 kV substations. Each station is assigned two planning variables: a capacity-load ratio and a storage power rating for absorbing reverse power. Current peak loads are projected to the planning year with a 5% annual growth rate over a five-year horizon. The projected load of station j, denoted by L_j^p, is used consistently in the capacity, reverse-power, and N-1 risk terms.

The pre-storage midday reverse injection at station j is denoted by r_j^0. After storage absorption, the residual reverse power is given by

$$\tilde{r}_j(P_j) = max\{0, r_j^0 - P_j\}. \tag{1}$$

Here, P_j denotes the storage power rating assigned to station j. $\tilde{r}_j(P_j)$ is the residual reverse power after storage absorption. The capacity-load ratio R_j is bounded within [1.2, 3.0], and the storage power rating P_j is bounded within [0, 6] MW. The regional capacity-load ratio is constrained by the planning band in (2),

$$1.8 \le \sum_j R_j L_j^p / (\sum_j L_j^p) \le 2.2\,. \tag{2}$$

This ratio is total planned capacity divided by the county planning-year peak. Residual reverse power is constrained at both station and upstream levels,

$$\begin{gathered} \tilde{r}_j(P_j) \le \beta R_j L_j^p \\ \eta \sum_j \tilde{r}_j(P_j) \le \beta R_{220} \delta \sum_j L_j^p\,. \end{gathered} \tag{3}$$

where β = 0.85 is the transformer reverse-loading limit, R_{220} = 1.8 the upper-level capacity-load ratio, δ = 0.85 the upstream coincidence factor, and η = 1.0 the reverse coincidence factor. The first inequality limits station-level residual reverse power, while the second limits aggregated reverse power at the upstream interface.

The optimization minimizes two conflicting objectives: annualized cost and N-1 supply risk,

$$\begin{gathered} \min\,(f_1, f_2) \\ f_1 = \sum_j (c_R R_j L_j^p + c_P P_j + C_j^{loss} + C_j^{om}) \\ f_2 = \sum_j \mathrm{EENS}_j(R_j, T_j, L_j^{\mathrm{p}})\,. \end{gathered} \tag{4}$$

Here, f_1 is the annualized cost objective and f_2 is the N-1 risk objective. The parameters c_R and c_P are annualized unit costs, while C_j^{loss} and C_j^{om} denote annualized loss and O&M costs. The N-1 risk f_2 is the total EENS,

$$EENS_j = \sum_{s \in \Omega_j} p_s \, h_s max\{0, L_j^p - S_{j,s}\} \; . \qquad (5)$$

In (5), Ω_j is the outage-state set, p_s and h_s are the probability and duration of state s, and $S_{j,s}$ is the available supply determined by transformer capacity and fixed tie-line support.

Storage is considered only for reverse-power absorption and is not credited as firm N-1 backup. The resulting optimization is subject to the variable bounds, the regional ratio band, and the reverse-power constraints.

B. EPG-NSGA-II Procedure

EPG-NSGA-II retains the NSGA-II backbone [6] and introduces warm-start sampling and repair projection before objective evaluation. Warm-start sampling seeds a near-feasible initial population; repair projection is applied after every variation step.

Algorithm 1 EPG-NSGA-II

Input: D, bounds, ratio band, reverse-power parameters, N, G.
Output: feasible nondominated set A.

```
P ← Warm-start sampling(D, N)
P ← Repair projection(P)
Evaluate f1 and f2 for all individuals in P
for t = 1 to G do
   Q ← Variation(P)
   Q ← Repair projection(Q)
   Evaluate f1 and f2 for all individuals in Q
   U ← P ∪ Q
   P ← nondominated sort + crowding select (U, N)
end for
A ← feasible nondominated individuals in P
return A
```

公式？或者更形式化、规范化的表达?

WarmStartSampling samples ratios within engineering bounds, projects the aggregate ratio into the planning band, and allocates storage to stations with higher reverse-power pressure.

公式？或者更形式化、规范化的表达?

RepairProjection clips variables, repairs the aggregate-ratio band, checks station-level and upstream reverse-power constraints, and performs a final feasibility check.

III. EXPERIMENTS

A. Setup

The benchmark is a public-data-informed synthetic county constructed from scaled IEEE 33-bus feeders [7] and PVGIS irradiance [8]. We test four county sizes, K=10, 20, 30, and 40, with population 100 and three seeds per setting. Plain NSGA-II is run for 800 generations as a longer-budget reference, while EPG-NSGA-II is run for 400 generations.

B. Efficiency and Ablation

TABLE I. NORMALIZED HYPERVOLUME RESULTS

Method	K=10	K=20	K=30	K=40
EPG-NSGA-II	1.000	1.000	1.000	1.000
Repair-only	0.997	0.998	0.999	1.000
NSGA-II	0.993	0.993	0.991	0.954

Table I reports normalized converged HV across county sizes. Repair-only remains within 0.3% of the full method, suggesting that repair projection is important for converged quality in this setting. Plain NSGA-II shows a larger gap as K grows, reaching 4.6% at K=40. At K=40, EPG-NSGA-II reaches a comparable HV with fewer generations under the fixed-population setting.

C. County-Level Application at K=20

At K=20, the knee solution has a regional capacity-load ratio of 1.80, an annualized cost index equivalent to 188.82 million CNY/yr, and EENS of 114 MWh/yr. PV-heavy stations receive storage, while low-ratio stations rely more on tie-line support (Fig. 1). Relative to the uniform R=2.0 baseline, the elastic configuration reduces the annualized cost index by 13.1% under equal EENS in this synthetic case.

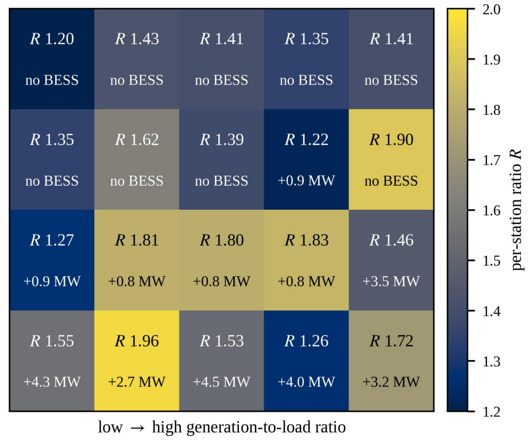

Fig. 1. Elastic planning configuration for the synthetic county case.

IV. CONCLUSION

EPG-NSGA-II combines feasibility-aware initialization and repair projection for elastic capacity-load ratio planning under high PV penetration. The synthetic cases show a lower cost index under equal EENS and improved search efficiency compared with plain NSGA-II. Future work will consider real-grid calibration, energy-limited storage, and statistical testing.

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